# Incident Investigation Report

Sysmon / Wazuh Telemetry Analysis — Isolated Lab Environment

Date of Activity: 2026-10-03 | Host: DESKTOP-5OG7C28 (Windows_VM)

## 1. Executive Summary

Analysis of the supplied Sysmon process creation, file creation, service, and related Windows event telemetry, together with the corresponding Wazuh alerts, shows a short sequence of user-initiated activity on host DESKTOP-5OG7C28 under the account Darragh Casey. The dominant observed behaviour is local account and credential discovery. A PowerShell process (PID 5664) was used interactively to execute hostname.exe and whoami.exe multiple times, after which Notepad was used to create a script file named Test_2.ps1 inside a desktop folder explicitly named T1087.001. Shortly thereafter the same PowerShell process spawned cmd.exe which executed a chained discovery command line containing net user, dir c:\Users\, cmdkey.exe /list, and net localgroup. These actions produced multiple Wazuh alerts mapped to MITRE ATT&CK; T1087 (Account Discovery) and T1059.003 (Windows Command Shell).

Additional alerts (dllhost.exe process injection, BITS service start-type changes, and an executable written under SoftwareDistribution) were examined against the underlying events. The dllhost.exe instance matches the legitimate COM Surrogate binary and has no supporting injection indicators in the supplied data. The BITS changes and the SoftwareDistribution executable are temporally and contextually associated with legitimate Windows Update / Windows Defender definition update activity. No evidence of successful lateral movement, persistence beyond normal update cleanup, or confirmed malicious payload execution was present in the supplied telemetry.

Overall confidence in the identification of intentional local account discovery activity is high, given the explicit folder naming, the script creation, and the subsequent execution of matching discovery commands. Confidence that the remaining alerts represent benign system activity is medium-to-high.

## 2. Scope of Activity

The supplied evidence covers approximately ten minutes of activity (approximately 16:21–16:31 UTC on 2026-10-03) on a single Windows host named DESKTOP-5OG7C28 (agent Windows_VM, IP 192.168.232.130). The data consists of 15 Wazuh alerts and 160 archived Windows event-channel records, the majority of which are Sysmon Event ID 1 (Process Create) records. Activity is limited to process execution, file creation by Notepad, two BITS service start-type changes, Windows Update / Defender-related process and file activity, one outbound network connection from OneDriveStandaloneUpdater.exe, and numerous Credential Manager access events (Security Event ID 5379). No evidence of remote logons, network shares, or multi-host activity appears in the data.

## 3. Detailed Investigation

### 3.1 Interactive PowerShell and Discovery Commands (Observed)

A PowerShell process (C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe, PID 5664, integrity High, user DESKTOP-5OG7C28\Darragh Casey) is the parent of numerous child processes throughout the window. The process creation event for PID 5664 itself is not present in the supplied archive, indicating it was already running when the capture began. From this process the following were executed (all High integrity, same user):

• Multiple instances of hostname.exe and whoami.exe between 16:21:19 and 16:26:14 UTC. These are consistent with interactive command testing or basic host/user enumeration.

• At 16:26:15.192 UTC, cmd.exe (PID 11280) with the exact command line:

```
"cmd.exe" /c net user & dir c:\Users\ & cmdkey.exe /list & net localgroup "Users" & net localgroup
```

Child processes of this cmd.exe include net.exe / net1.exe (net user, net localgroup "Users", net localgroup) and cmdkey.exe /list. These commands enumerate local user accounts, local groups, the contents of C:\Users, and stored credentials. This is the primary activity that generated the cluster of Wazuh rules 92052, 92036, 92031 and 92032.

### 3.2 Script File Creation (Observed)

At 16:23:47.871 UTC, Notepad.exe (Microsoft.WindowsNotepad package) created the file C:\Users\Darragh Casey\Desktop\T1087.001\Test_2.ps1. A second FileCreate event for the same path occurred one second later (likely a write/close or duplicate logging). The parent folder is explicitly named with the MITRE ATT&CK; technique identifier T1087.001 (Account Discovery: Local Account). No Sysmon process-creation event showing powershell.exe or pwsh.exe executing Test_2.ps1 appears in the supplied data; the discovery commands were instead launched via the already-running PowerShell process spawning cmd.exe. Inference: the script was prepared in the MITRE-named folder and its contents (or equivalent commands) were subsequently run interactively.

### 3.3 dllhost.exe Alert (Observed + Assessment)

At 16:22:44.804 UTC, dllhost.exe (PID 10552, Medium integrity, user Darragh Casey) was created with command line "C:\WINDOWS\system32\DllHost.exe" /Processid:{9F156763-7844-4DC4-B2B1-901F640F5155}. Parent process information is absent (ParentProcessId 952, ParentImage blank), which is typical for COM surrogate activation. The reported hashes match the legitimate Microsoft dllhost.exe binary. Wazuh rule 61638 (level 12) mapped this to T1055 Process Injection. No additional Sysmon events (module loads, process access, remote thread creation, etc.) supporting injection were present in the supplied telemetry. Conclusion: the alert is based solely on the presence of dllhost.exe; the underlying evidence does not support malicious process injection.

### 3.4 BITS Service Changes and Windows Update Activity (Observed)

Service Control Manager Event ID 7040 recorded that the Background Intelligent Transfer Service (BITS) start type changed from demand start to auto start at 16:23:00 UTC, then back to demand start at 16:25:05 UTC. Concurrent activity includes UsoClient.exe StartWork, MoUsoCoreWorker.exe, and later the creation of C:\Windows\SoftwareDistribution\Download\Install\AM_Delta_Patch_1.459.518.0.exe by svchost.exe (rule 92217). This is consistent with normal Windows Update / Windows Defender definition update behaviour. The temporary BITS start-type change is a known pattern during update operations and does not, by itself, indicate persistence.

### 3.5 Other Activity Assessed as Unrelated or Benign

• A RunOnce registry value set by EdgeWebView setup.exe (Sysmon EID 13, ruleName T1060,RunKey) is a standard cleanup entry left by the installer.

• One Sysmon network connection (EID 3) from OneDriveStandaloneUpdater.exe to 52.123.128.14:443.

• Numerous Security Event ID 5379 (Credential Manager credentials were read) events, several of which align temporally with the cmdkey.exe /list execution.

These events do not alter the primary finding of local account discovery.

## 4. Evidence Analysis

Key supporting records (timestamps are UTC as recorded in Sysmon eventdata):

• Process Create (Sysmon EID 1) 2026-10-03 16:26:15.192 — cmd.exe PID 11280, parent powershell.exe PID 5664, command line containing the chained net / dir / cmdkey discovery commands. Integrity High, user Darragh Casey.

• Process Create events for net.exe / net1.exe / cmdkey.exe as children of the above cmd.exe, confirming execution of each component.

• File Create (Sysmon EID 11) 2026-10-03 16:23:47.871 and 16:23:48.051 — Notepad.exe writing Test_2.ps1 under Desktop\T1087.001.

• Multiple Process Create events for hostname.exe and whoami.exe with parent powershell.exe PID 5664 between 16:21 and 16:26.

• Process Create 2026-10-03 16:22:44.804 — dllhost.exe with the noted CLSID; hashes consistent with legitimate binary; no supporting injection telemetry.

• Service Control Manager 7040 events at 16:23:00 and 16:25:05 documenting the BITS start-type toggle.

• File Create 2026-10-03 16:27:15.554 — AM_Delta_Patch_1.459.518.0.exe under SoftwareDistribution by svchost.exe.

Wazuh alerts 92031, 92032, 92036 and 92052 correctly identify the discovery command sequence. Alert 61638 (dllhost) and alert 92217 (executable in SoftwareDistribution) do not have corroborating malicious indicators in the underlying events.

## 5. Timeline of Events

16:21:19–16:26:14 UTC — Repeated hostname.exe and whoami.exe executions from PowerShell PID 5664 (user Darragh Casey, High integrity).

16:22:44.804 UTC — dllhost.exe launched as COM surrogate (CLSID {9F156763-7844-4DC4-B2B1-901F640F5155}).

16:23:00 UTC — BITS service start type changed from demand to auto.

16:23:47.871 UTC — Notepad creates Desktop\T1087.001\Test_2.ps1.

16:25:05 UTC — BITS service start type changed from auto back to demand.

16:26:15.192 UTC — PowerShell PID 5664 spawns cmd.exe with chained discovery command line (net user, dir c:\Users, cmdkey /list, net localgroup).

16:26:15.264–16:26:15.459 UTC — Child processes net.exe, net1.exe and cmdkey.exe execute the individual discovery commands.

16:27:15.554 UTC — Defender definition delta patch written under SoftwareDistribution (legitimate update activity).

16:29:49 UTC — EdgeWebView installer writes a RunOnce cleanup key (legitimate).

## 6. MITRE ATT&CK; Mapping

T1087 / T1087.001 — Account Discovery (Local Account). Supported by the explicit execution of net user, net localgroup, and dir c:\Users, plus the preparatory creation of a script inside a folder named T1087.001. Multiple Wazuh alerts correctly mapped this activity.

T1059.003 — Windows Command Shell. Supported by the cmd.exe /c invocation of the discovery chain from PowerShell.

T1059 — Command and Scripting Interpreter. Supported by the interactive PowerShell process and the creation of a .ps1 file.

T1555.004 / Credential Access (related) — The inclusion of cmdkey.exe /list constitutes credential enumeration; this is also reflected in associated Security 5379 events.

T1055 — Process Injection. Mapped by Wazuh solely on the basis of the dllhost.exe process creation. No supporting evidence of injection is present; mapping is not supported by the telemetry.

T1570 — Lateral Tool Transfer. Mapped by Wazuh on the SoftwareDistribution executable write. The file is a known Windows Defender delta patch written by svchost; mapping is not supported.

## 7. Overall Investigation Conclusion

The supplied telemetry documents intentional local account and credential discovery activity performed by the interactive user Darragh Casey on an isolated laboratory host. The activity was prepared by creating a PowerShell script inside a desktop folder explicitly named for the corresponding MITRE technique (T1087.001) and was subsequently executed via a chained cmd.exe command line launched from an existing PowerShell process. This sequence is the clear, high-confidence finding of the investigation.

The remaining alerts were evaluated against their underlying events. The dllhost.exe alert lacks any corroborating indicators of process injection. The BITS service changes and the executable written under SoftwareDistribution are consistent with routine Windows Update and Defender definition update activity. Other registry and network events are attributable to normal application behaviour (EdgeWebView, OneDrive).

Observed facts are limited to the process, file, service and event records described above. Inferences (that the .ps1 contents were the source of the later discovery commands, and that the activity was a deliberate lab simulation) are drawn from the folder naming, the temporal proximity, and the isolated laboratory context stated in the tasking. Overall confidence in the primary conclusion of local account discovery is high. Confidence that the non-discovery alerts are benign is medium-to-high.

End of Report